

MOHANRAJ RAMKUMAR

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Professional Summary

AI Engineer with hands-on experience building production-grade Generative AI and ML systems using Python, LLMs, RAG, AI Agents, LangChain, LangGraph, FastAPI, Flask, Scikit-learn, TensorFlow, and PyTorch. Experienced in RAG pipeline design, model training and evaluation, agent orchestration via MCP, and ML pipeline automation with Airflow and MLflow. Hands-on with Databricks, Azure, AWS, Docker, Kubernetes, and cloud-based AI solutions. Databricks Certified Generative AI Engineer Associate.

Technical Skills

Programming: Python, JavaScript, HTML, CSS

Backend / APIs: FastAPI, Flask, Django, REST APIs, Microservices, Postman, Swagger

Generative AI / LLM: Generative AI, LLMs, RAG, AI Agents, Prompt Engineering, LangChain, LangGraph, Model Context Protocol (MCP), Ollama, GPT-4, Gemini, Hugging Face, LLM Fine-Tuning

ML / DL: Machine Learning, Deep Learning, NLP, Scikit-learn, TensorFlow, PyTorch, Pandas, NumPy, Model Evaluation, Hallucination, Toxicity, Bias, Drift Monitoring

Vector Databases: ChromaDB, Vector Search, Embedding Pipelines

Pipelines / MLOps: Airflow, MLflow, Airbyte, CI/CD Pipelines

Cloud / Data: Azure, Azure OpenAI, Databricks, Unity Catalog, Databricks SQL Warehouses, AWS Lambda, AWS SQS, Azure SQL, MySQL, MongoDB

DevOps / Tools: Docker, Kubernetes, Git, GitHub, GitHub Copilot, Windsurf, Amazon Bedrock, Claude Code, Streamlit, Vercel

Professional Experience

Generative AI Developer – Data Pattern

Jun 2025 – Aug 2026

- Built and deployed a full-stack conversational analytics platform using Python, Flask, React, LangChain, and Databricks LLMs, converting natural-language questions into SQL queries and interactive visualizations.
- Designed and implemented RAG pipelines using ChromaDB for document embedding, retrieval, and context-aware response generation; applied Scikit-learn, TensorFlow, and PyTorch for model training and evaluation.
- Orchestrated agent-based task execution using Model Context Protocol (MCP) and LangGraph; automated ML/data pipelines using Airflow and tracked experiments using MLflow.
- Developed production-ready REST APIs and backend services, testing and debugging using Postman and Swagger; built a data-profiling engine orchestrating 40+ Airbyte connectors through Docker.
- Built an LLM-powered metadata generation system using Databricks Unity Catalog, Ollama, and Gemini to generate column descriptions, tags, and sensitivity classifications.
- Designed LLM and ML/DL evaluation and monitoring engines covering completeness, accuracy, toxicity, hallucination, bias, relevancy, and drift; created synthetic evaluation datasets and automated testing workflows.
- Built an LLM-driven ETL toolkit generating production-ready Python pipelines from mapping documents, plus an AWS Lambda telemetry agent for compute, GPU, and bandwidth monitoring.

Data Science Intern – Postulate

May 2024 – Jun 2024

- Performed data cleaning, EDA, statistical analysis, and visualization using Python, NumPy, Pandas, Matplotlib, and Scikit-learn.

Key Projects

RAG-Based HR Chatbot: Built a RAG chatbot using Ollama, LangChain, and ChromaDB to retrieve document context and generate context-aware responses.

AI-Based Data Analysis Assistant: Built an AI assistant using GPT-4 and NLP to process CSV files, answer natural-language questions, and generate visualizations.

Achievements

Finalist – RBI HaRBInger 2024, Reserve Bank of India, FinTech Track | Finalist – Beyond Abstraction 2024, Blockchain Track | Participant – GDG Hackathon 2024 and Smart India Hackathon 2024

Certifications

Databricks Certified Generative AI Engineer Associate | Oracle Cloud Data Management Foundations | Oracle Cloud Infrastructure Foundations | Oracle Java Foundations | Infosys Springboard: Machine Learning, Git/GitHub, MongoDB NoSQL

Education

B.Sc. Artificial Intelligence – Rathinam College of Arts and Science, Coimbatore
CGPA: 8.0

2023 – 2026